

Lecture 1: Basic Concepts

EE201 – Circuit Theory I

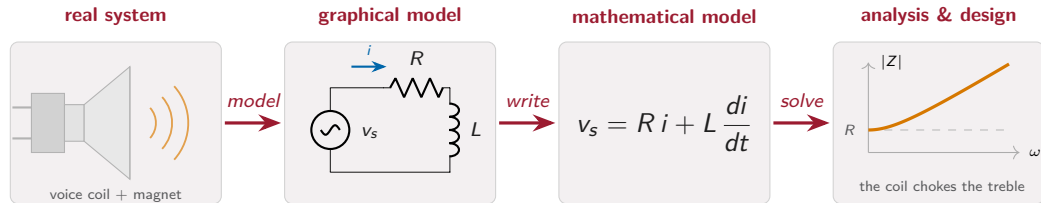
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charge ■ current ■ voltage ■ power ■

these colours mean the same thing on every slide and in the notes

Circuit theory is not about wires

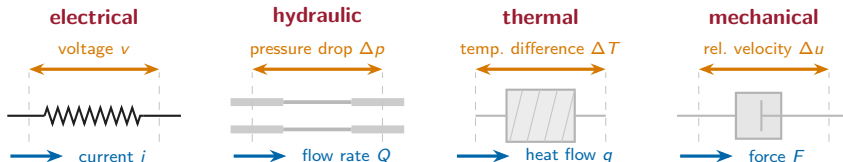


A real loudspeaker also has a cone, a suspension, a mass, a room around it.
We keep **two numbers**, R and L , and throw the rest away.

"All models are wrong, but some are useful."

George E. P. Box, 1976

The same pipeline, other domains



Across: needs *two* points

A difference. The second point is usually left unsaid because a reference was agreed on long ago — sea level, atmospheric pressure. A circuit has none, so we pick one: *ground*.

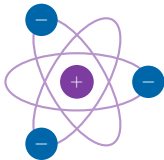
Through: needs *one*

Measured at a place, with a direction — and no reference: zero current means the same to everyone.

Charge

Definition

Charge is the property of matter that makes it feel a force in an electromagnetic field. Symbol q , unit **coulomb (C)**.



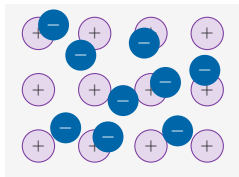
one atom: as many $-$ as $+$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$1 \text{ C} \approx 6.24 \times 10^{18} \text{ electron charges}$$

- electron: $-e$ proton: $+e$
- **Charge is conserved** — never created or destroyed, only moved. (This will become KCL.)

A wire is already full of charge



copper: fixed + ions, electrons
free to move *within* the metal



a pipe **already full**: raise the pressure
at one end, the far end flows *at once*

1 cm³ of copper contains about

8.5×10^{22} free electrons $\approx$ 13 600 C

of *mobile* charge — free to move **within**
the metal, not to leave it.

And its net charge is **zero**: every electron
that broke loose left its + ion behind.

So what is a current?

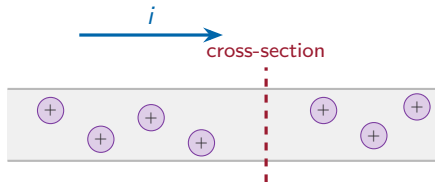
Charge that is *already there*, **drifting** — not
charge being poured in. Nothing is used up:
what flows in at one end flows out at the other.

Current: charge in motion

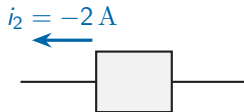
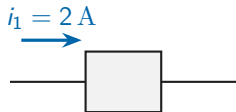
Definition

$$i(t) = \frac{dq(t)}{dt} \quad \Longleftrightarrow \quad q(t) = q(t_0) + \int_{t_0}^t i(\tau) d\tau$$

Unit: **ampere (A)**, $1 \text{ A} = 1 \text{ C/s}$.



The arrow is a choice, not a fact



Same physical situation. The arrow says which way we call positive; the sign of the number says which way the charge really goes.

A negative answer is not a mistake — it is the model telling you something.

Two numbers worth knowing

How much is a coulomb?

A 4000 mAh phone battery:

$$Q = 4 \text{ A} \times 3600 \text{ s} = 14\,400 \text{ C}$$

$$\approx 9 \times 10^{22} \text{ elementary charges}$$

Charge comes in whole electrons — but with 10^{22} of them we can treat $q(t)$ as a **smooth** quantity, which is what lets us write dq/dt at all.

How fast do the charges move?

1 A in a 1 mm^2 copper wire:

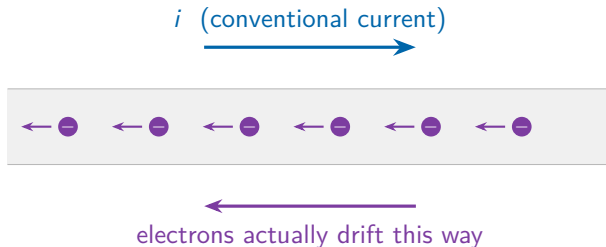
each electron drifts at about

$$0.07 \text{ mm/s} \approx 1 \text{ metre in 4 hours}$$

Yet the lamp lights the instant you flip the switch. The wire is **already full**, so every electron starts moving together — you never wait for one to travel from the switch to the lamp.

We come back to that gap when we define a *lumped* circuit.

Conventional current

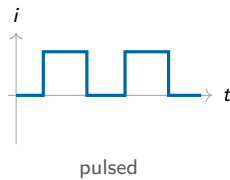
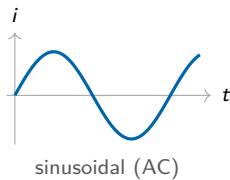
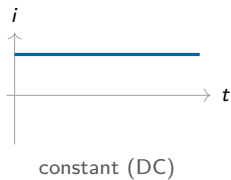


One electron moving left carries as much charge to the right
as one positive charge moving right.

Just as well: in a semiconductor the carriers are electrons *and* holes, and in a battery they are ions.

Notation

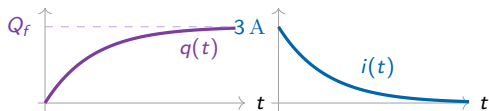
may vary in time **constant**
 $i(t), v(t), q(t), p(t)$ I, V, Q, P



Example: charging a phone

The last part of a charge cycle: the charge *stored in the battery* approaches its full value, $q(t) = Q_f (1 - e^{-t/\tau})$, with $Q_f = 3600 \text{ C}$ and $\tau = 20 \text{ min}$.

$$i(t) = \frac{dq}{dt} = \frac{Q_f}{\tau} e^{-t/\tau}$$

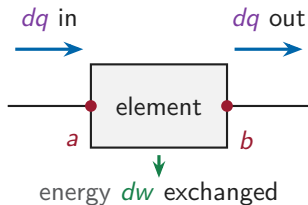


- $i(0) = 3600/1200 = 3 \text{ A}$
- after 20 min: $i = 3/e \approx 1.1 \text{ A}$
- the fuller it gets, the slower it fills

Charge is what accumulates; current is how fast it accumulates.

Voltage: why the charge moves

Charge does not move by itself. Pushing it through an element **costs or releases energy** — and that is what voltage measures.



Definition

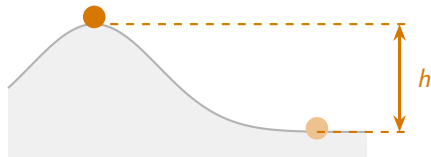
$$v_{ab} = \frac{dw}{dq} \quad \text{unit: } \mathbf{volt\ (V)}, \quad 1\text{ V} = 1\text{ J/C}$$

the energy exchanged *per unit of charge* carried from *a* to *b*.

What that definition does — and does not — say

- It is a **ratio**. Twice the charge, twice the energy — same v_{ab} . Voltage belongs to the **pair of points**, not to the charge.
- It does **not need a current**. A 1.5 V battery on the shelf has 1.5 V across it at $i = 0$ — and still 1.5 V while it drives 2 A. Voltage is defined either way: it does not depend on whether charge happens to be flowing.
- **Order matters**: $v_{ab} = -v_{ba}$. There is no “voltage at a point” until we fix a reference.
- The energy per charge depends on the two endpoints, **not on the path**. So “the voltage between a and b ” is a **single, definite number** — take whichever path you like and you get the same answer. Later this becomes KVL.

Voltage is an electrical height



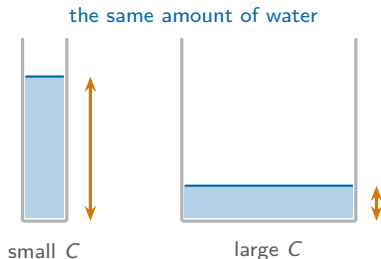
energy per unit *mass* to lift: gh

- Only the height **difference** matters.
- “Sea level” is a **choice** — and changing it changes no physics.
- A heavier ball gains more energy, but energy *per kilogram* is the same: that is why gh , like voltage, is a property of the two places.

Voltage is the electrical version of this, with charge in place of mass.

Charge is the amount, voltage is the level

$$q = C v$$

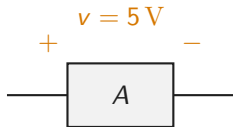


- **charge** — the amount poured in
- **voltage** — the level it reaches
- C — the cross-section of the container

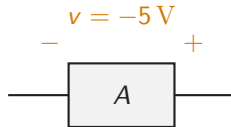
The **shape of the circuit** plays the part of the shape of the tank. We define C properly in Lecture 5.

Reference polarity

Exactly as for current, we **can** mark $+$ and $-$ before we know the answer.



left terminal 5 V above the right one



same element, *same* state

Same physical situation, two bookkeeping choices — exactly like the current arrow.

The first element: the resistor

We have defined the variables. To talk about a circuit we also need an *element* — a rule tying v and i together. Here is the simplest one.

Ohm's law

$$v = Ri$$

R in ohms (Ω), $1\ \Omega = 1\ \text{V}/\text{A}$



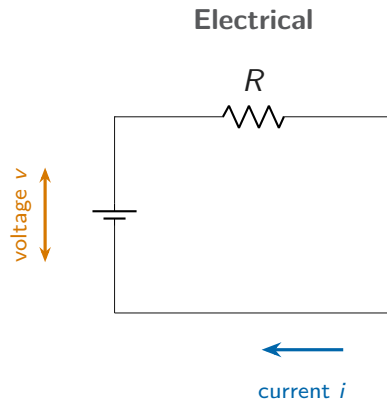
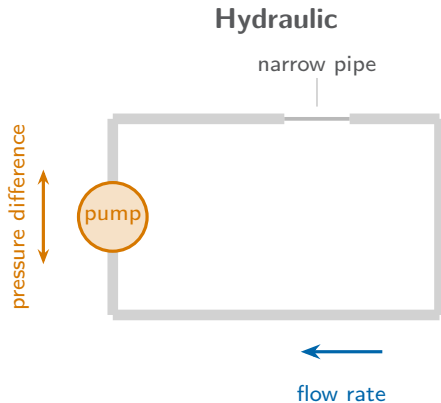
- **Algebraic, no memory** (*static*): the current *now* depends only on the voltage *now* — no derivatives, no history.
- **Geometry sets R** : for a uniform bar,

$$R = \rho \frac{L}{A}$$

long and thin $\Rightarrow$ large R ; short and fat $\Rightarrow$ small R . Exactly like a pipe.

The full family of elements — and the sign convention that goes with them — comes in Lecture 2.

The hydraulic analogy



The dictionary

Hydraulic		Electrical
pressure difference	$\leftrightarrow$	voltage v
volumetric flow rate	$\leftrightarrow$	current i
pipe restriction	$\leftrightarrow$	resistance R
pump	$\leftrightarrow$	voltage source

Useful for intuition, not for proofs — water is incompressible, charge is not.

Ground: the node we choose to call zero

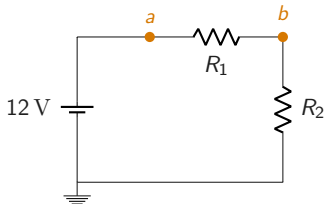
Reference node

Only differences matter, so we are free to declare one node to be 0 V. Then

$$V_{ab} = V_a - V_b.$$

Move the reference and every node voltage shifts by the same constant. No two-point voltage, current or power changes.

Pick the node that the most elements touch — your equations get shorter.



Three different things are called “ground”

Earth ground

A physical connection to the planet. Safety.

Chassis ground

The metal enclosure of the instrument.

Signal ground

The analysis reference. **This is ours.**

In EE201 we mean the third one.

In the laboratory, confusing them is how you blow a fuse.

Where $p = vi$ comes from

$$p(t) = \frac{dw}{dt} = \underbrace{\frac{dw}{dq}}_{v(t)} \cdot \underbrace{\frac{dq}{dt}}_{i(t)} = v(t) i(t)$$

Power

unit **watt (W)**, $1 \text{ W} = 1 \text{ J/s}$

Energy

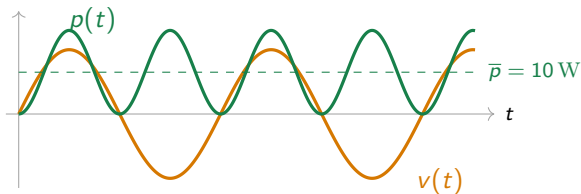
$$w = \int_{t_0}^t v(\tau) i(\tau) d\tau$$

$p = vi$ is **not a new definition** — it follows from defining v as energy per charge and i as charge per time.

Power that changes in time

A resistive element with $v(t) = 10 \cos \omega t$ V and $i(t) = 2 \cos \omega t$ A:

$$p(t) = vi = 20 \cos^2 \omega t = 10(1 + \cos 2\omega t) \text{ W}$$



- $p(t) \geq 0$: a resistor never gives energy back.
- The power **oscillates** at *twice* the frequency, around $\bar{p} = 10 \text{ W}$.

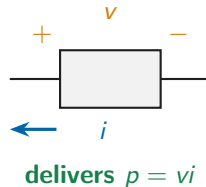
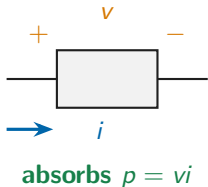
The passive sign convention

PSC

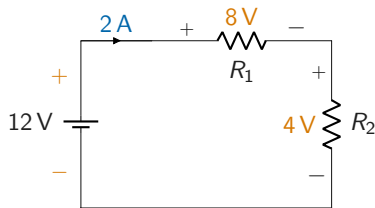
Draw the current arrow so that it **enters the + terminal**. Then

$$p = vi = \text{power absorbed by the element.}$$

$p > 0$: the element absorbs. $p < 0$: it delivers.



Example: a power balance



$$p_{R_1} = 8 \times 2 = +16 \text{ W}$$

$$p_{R_2} = 4 \times 2 = +8 \text{ W}$$

$$p_{\text{src}} = 12 \times (-2) = -24 \text{ W}$$

$$16 + 8 - 24 = 0 \checkmark$$

The source current *leaves* its + terminal —
hence -2 A .

In *any* circuit, at *every* instant, $\sum_k p_k(t) = 0$. If the powers in your solution do not cancel, your solution is wrong — **a free check, every time.**

What exactly is a circuit?

Elements interconnected so that charge can flow along **closed paths**.

Everything in EE201 also assumes the circuit is **lumped**:

Three things we assume

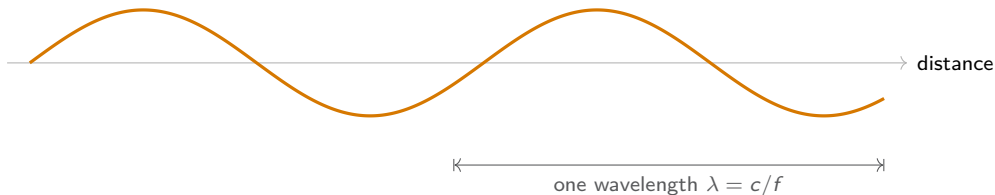
- 1 Nothing piles up.** Current out = current in, at every instant. Charge may *separate* inside — a capacitor does exactly that — but the element as a whole never gains charge.
- 2 Nothing leaks out.** No changing magnetic field through your loops: they are neither antennas nor transformers.
- 3 Nothing is late.** The circuit is much smaller than the wavelength, $d \ll \lambda = c/f$.

50 Hz: $\lambda = 6000$ km — any board is lumped.

2.4 GHz: $\lambda = 12.5$ cm — a 3 cm track is not.

Read the axis carefully

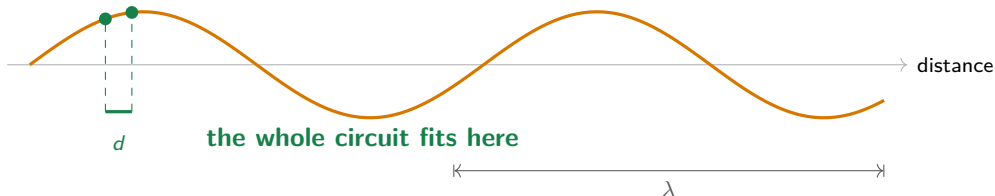
A signal does not appear everywhere at once — it **travels** along the circuit at about c . So picture the whole circuit **frozen at one instant**, and plot the voltage **along it**.



The horizontal axis is distance, not time. Every point of the circuit sits somewhere on this wave.

The question is whether the circuit is **short enough** that its two ends sit at the *same* place on it.

Short circuit, one number: $d \ll \lambda$

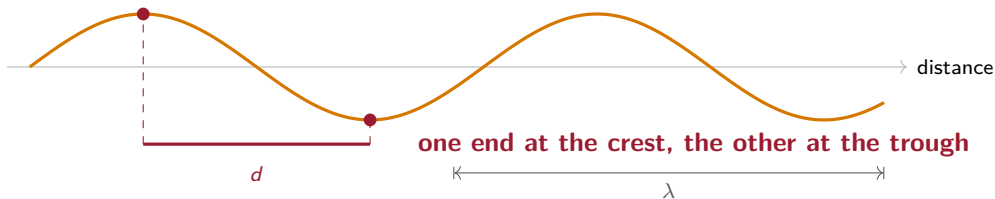


Both ends sit at practically the **same point** of the wave, so every point of the circuit reads **the same number** at that instant.

That is what lets us say “the voltage of a node” at all.

Mains, $f = 50 \text{ Hz}$: $\lambda = 6000 \text{ km}$. A 30 cm board is 10^{-7} of a wavelength. **Everything in EE201 lives here.**

... and when it is not: $d \sim \lambda$



Different values at the same moment. There is no single number to call “the node voltage”, so circuit theory no longer applies — you need the fields themselves: transmission lines, and Maxwell’s equations.

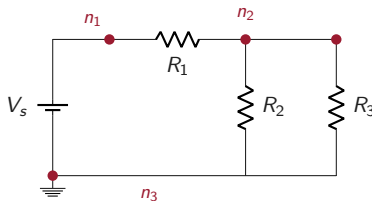
Wi-Fi, $f = 2.4 \text{ GHz}$: $\lambda = 12.5 \text{ cm}$.
A 3 cm trace is a *quarter* of a wavelength.

The same board can be lumped for one signal and not for another.

Terminals, nodes, branches, loops

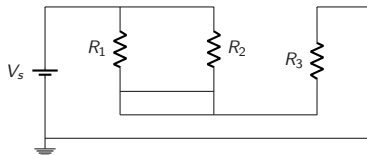
- **Terminal** — a point where an element connects. R, L, C: two. Transistor: three.
- **Node** — a connection point. Everything joined by ideal wire is **one** node, however long the wire.
- **Branch** — one element between two nodes.
- **Loop** — a closed path through no node twice.

$$\# \text{branches} = \# \text{loops} + \# \text{nodes} - 1$$

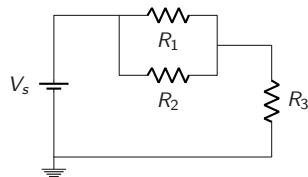


$$4 = 2 + 3 - 1$$

The same circuit, drawn twice



drawing (a)



drawing (b)

Same circuit. **Drawing is not topology.**

When a problem looks impossible, redraw it so that everything joined to the same node sits together — the series and parallel connections then show themselves.

Summary

Quantity	Symbol	Definition	Unit
charge	q	—	C
current	i	$i = dq/dt$	A = C/s
voltage	v	$v = dw/dq$	V = J/C
power	p	$p = vi$	W = J/s
energy	w	$w = \int p dt$	J

- References (arrow, polarity) are **choices**. The **sign** of the answer then tells you what is really happening — a minus is information, not an error.
- PSC: current into $+$ $\Rightarrow p = vi$ is absorbed, and $\sum_k p_k = 0$.
- Across vs through. Ground is a choice.
- Lumped $\Rightarrow$ ODEs. That is what makes this course possible.

Next lecture

Interconnection laws

Kirchhoff's current law (conservation of charge) and Kirchhoff's voltage law (conservation of energy), then the branch relations of the basic elements.

Questions?

Notes for this lecture: see the course repository.